Partner-Specific Affective Precision in Social Active Inference

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Abstract. Social agents must infer what their partners are likely to do and decide how strongly those inferences should guide action. Active inference separates these problems: beliefs about hidden social states support a policy posterior, while policy precision controls how decisively the agent acts on that posterior. Hesp et al. [10] give this precision psychological content by formalizing valence as inference over expected action precision, or subjective model fitness. On this view, affect is not a direct readout of reward; it is a higher-order estimate of how well the agent's model is currently supporting action. We ask how this mechanism behaves when model fitness is local to particular social partners. A focal active-inference agent plays a repeated multi-partner trust game, maintaining partner-specific beliefs about each partner and a partner-specific precision estimate updated from prediction error about that partner's behavior. The simulations ask three questions: whether affective precision tracks predictive reliability rather than payoff, whether it changes action selection independently of belief accuracy, and when it helps or harms social behavior. The results support a conditional interpretation. Partner-local precision tracks predictive reliability more strongly than realized reward ($|\text{corr}| = 0.701$ vs. 0.419), indicating a metacognitive signal about model fit rather than cached social value. In decision-open graded regimes it improves payoff and reduces policy entropy even when partner-belief accuracy is matched, placing the effect at the policy-selection stage. In partner-choice regimes it shifts engagement toward more predictable partners before aggregate payoff separates. Under abrupt behavioral change, however, the same mechanism can reduce performance by sharpening action around a stale partner model. Partner-local affective precision therefore appears useful when social regularities persist long enough to support confident action, but risky when precision changes faster than partner beliefs can be revised.

Keywords: Active inference · Affect · Trust games · Social cognition · Precision

1 Introduction

Social agents face a problem that goes beyond accurate inference. They must learn what their partners are likely to do, but they must also decide how strongly those social beliefs should guide action. A well-calibrated belief that one partner defects roughly 80% of the time supports a different degree of commitment than a diffuse posterior over a newly encountered partner, even if both beliefs are represented by the same inference machinery. This calibration matters because social behavior is costly in both directions: over-committing to an uncertain partner model makes action brittle, while under-committing to a well-supported one leaves useful evidence idle.

Active inference frames perception and action as joint minimization of variational free energy under a generative model [4, 5, 13]. Agents infer the hidden causes of their observations via approximate Bayesian inference and evaluate candidate policies by their expected free energy—a quantity that balances preference satisfaction against epistemic value [6, 3]. The resulting policy posterior is then sharpened or broadened by a precision parameter γ: high precision concentrates action on the most-preferred policy; low precision leaves the distribution flatter, preserving room for exploration or caution.

Hesp et al. [10] extend this machinery to account for affective valence. Their key move is to treat precision not as a fixed hyperparameter but as a state the agent actively infers. The agent maintains a higher-order belief β over its expected action precision, representing how reliably its generative model is currently supporting action; Hesp et al. call this subjective model fitness. This makes affect a second-order quantity: valence emerges from inference over beliefs about beliefs. The updating signal, affective charge ϕ, is a signed and bounded prediction-error proxy: it rises when the model performs better than a prior fitness baseline and falls when it performs worse. The critical feature is that ϕ tracks *changes* in model fitness rather than reward itself. An agent can therefore have positive affective charge toward a reliably hostile partner when that hostility is well predicted, even though the expected outcome remains poor.

Multi-partner social settings add an important locality problem. A person can understand one partner well, remain uncertain about another, and be actively revising beliefs about a third. A single precision estimate collapses these situations into one global sense of confidence. That collapse is computationally convenient, but it obscures a social distinction that matters for action: confidence should often be directed toward the relationship whose model has earned it. We therefore extend affective precision to the partner level. Each partner has its own belief state and its own model-fitness estimate, allowing precision to modulate action locally rather than globally.

We study this extension in a repeated multi-partner trust game [2, 11]. The setting is useful because the agent faces a genuine social inference problem: partner types and dispositions are hidden and must be inferred from behavioral evidence. It also allows us to vary the decision regime. In binary settings, policy preferences can become nearly deterministic, leaving little room for precision to matter. In graded settings, multiple investment levels remain behaviorally plausible, making policy precision a visible part of action selection. This contrast lets us ask when partner-local affective precision changes behavior, not merely whether it can be added to the model.

Three questions organize the analysis:

1. Does the partner-local precision signal track realized reward or predictive reliability?
2. Does it change how confidently social beliefs drive action, independently of whether those beliefs are more accurate?
3. Under what conditions does it help, and what exposes its limits?

Together, these questions frame the contribution as a computational analysis rather than a claim that partner-local affect is universally beneficial. Across binary and graded regimes, partner-choice conditions, stable partners, abrupt betrayal, and gradual change, the simulations support three linked conclusions. First, the precision signal is more closely tied to predictive reliability than to realized reward, consistent with an interpretation as model fitness rather than cached social value. Second, when the policy space leaves room for precision to matter, partner-local affect changes behavior through action confidence rather than through better partner beliefs. Third, the same mechanism can become maladaptive when a partner changes faster than the belief state can recalibrate. The account therefore identifies affective precision as a relationship-local deployment signal: useful for acting on validated social models, but limited by the temporal stability of those models.

2 Methods

2.1 Trust game

A focal active-inference agent plays a repeated trust game [2, 11] with four partners over an episode of up to 200 rounds. Each partner has a latent *type* (cooperator, reciprocator, exploiter, or random) and a latent *disposition* toward the agent (trusting, neutral, or hostile). Neither is directly observable; the agent must infer them from behavioral evidence.

In the binary game both players choose to cooperate or defect on each round. Table 1 gives the focal agent's payoffs. The key feature is that defecting is tempting regardless of what the partner does, while mutual cooperation is collectively optimal—a standard social dilemma. In the graded regime, the binary choice is replaced by a discretized investment level; the agent decides how much to commit to the interaction rather than whether to commit at all. This opens the policy space substantially: rather than choosing between two actions separated by a large expected free energy gap, the agent distributes probability mass across several investment levels, so the precision parameter has room to shift which level is chosen. A partner-choice regime adds a further decision: at the start of each round the agent selects which partner to engage, turning approach and avoidance into explicit policy outputs.

2.2 Per-partner generative models

Rather than a single pooled social model, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k. This allows evidence about each

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates	Partner defects
Agent cooperates	3	−1
Agent defects	5	1

partner to accumulate independently and—crucially for the analysis below—allows the degree to which each model is acted upon to differ. For partner k the state space, observation space, and action space are:

$$s_k = (s_k^{\text{type}},\ s_k^{\text{disp}}) \qquad \text{(hidden: partner type and current disposition)} \tag{1}$$

$$o_k = (o_k^{\text{act}},\ o_k^{\text{pay}}) \qquad \text{(observed: partner response and payoff)} \tag{2}$$

$$a \in \mathcal{A} \qquad \text{(cooperate/defect or investment level)} \tag{3}$$

The model $\mathcal{M}_k = \{A_k^{\text{act}}, A_k^{\text{pay}}, B_k, C_k, D_k, E_k\}$ encodes: A_k^{act}, the likelihood that a partner of type s^{type} and disposition s^{disp} produces response o^{act}; A_k^{pay}, the likelihood of payoff given the agent's action and the partner's response; B_k, transition dynamics across rounds; D_k, prior beliefs over type and disposition; C_k, preferred payoff observations; and E_k, policy priors. Payoff is an observation modality rather than an external reward signal: it is predicted and updated within the same inference process as the partner's actions. State and policy inference use the official `pymdp` discrete active-inference implementation [9, 3].

2.3 Partner-local affective precision

In the Hesp et al. [10] framework, the agent maintains a higher-order state β representing expected action precision—its current estimate of how reliably the generative model is supporting confident action. An *affective charge* signal ϕ updates β round-by-round: ϕ is positive when the model predicts better than a prior fitness baseline, and negative when it predicts worse. This makes ϕ a derivative of model fitness rather than a static accuracy measure. The key theoretical point is that an agent can have high β toward a reliably hostile partner (good predictions, confident avoidance) and low β toward an erratic but rewarding one (poor predictions, cautious action) because the signal tracks how well the model is working, not how much value the partner delivers.

We extend this to the per-partner setting. After interacting with partner k, the agent computes a bounded prediction-error proxy on the partner's action:

$$\epsilon_k = 1 - P\Big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \;\Big|\; q(s_k^{\text{type}}, s_k^{\text{disp}})\Big)\,. \tag{4}$$

We use partner *action*, not payoff, because action is the most direct test of type-by-disposition predictive fit. Payoff can surprise even when the partner's behavior was perfectly predicted (e.g., a reliably defecting partner is never surprising, even though payoff is low). The proxy is bounded $[0, 1]$ rather than log-surprisal;

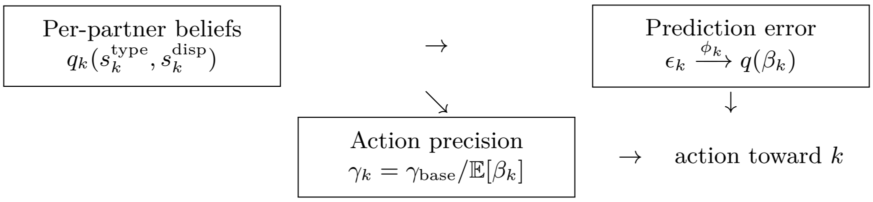

Fig. 1: Partner-local affective precision. Each partner has an independent belief state and precision estimate. Prediction error updates β_k, which sets how sharply future policies involving that partner are expressed in action.

this keeps charge ϕ_k in a range compatible with the update rule and preserves interpretability as a proportion of unexplained behavior.

The affective charge is:

$$\phi_k = \alpha\big(\sigma_0^2 - \epsilon_k^2\big)\,, \tag{5}$$

where σ_0^2 is a prior fitness baseline and α is a scaling constant. Positive charge means the model exceeded its prior reliability; negative charge means it fell short. A categorical posterior $q(\beta_k)$ over discrete precision levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$ is updated each round via a persistent Bayesian update weighted by ϕ_k. The expected precision then sets the softmax temperature for future decisions involving partner k:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{6}$$

When $\mathbb{E}[\beta_k]$ is low (the model has been consistently reliable), γ_k is large and the policy posterior is sharp. When $\mathbb{E}[\beta_k]$ is high (the model has been unreliable), γ_k is small and the posterior is diffuse. Note that β_k here plays the role of precision volatility: higher β means broader, less committed policy distributions. The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state inside the same variational model; this is a simplification relative to the full hierarchical treatment in Hesp et al. [10] and is discussed in Section 4.

Tracked-only condition. To isolate whether any behavioral effect of the precision mechanism runs specifically through action sharpening—rather than through some indirect influence on belief quality—we include a *tracked-only* variant. This variant retains the β_k update in full but fixes $\gamma_k = \gamma_{\text{base}}$ for all partners at all times, severing the link from the precision estimate to the softmax temperature. The β_k posterior can evolve normally and is available for analysis, but it never changes how policies are selected. If the tracked-only agent behaves like the no-affect agent, while full precision modulation differs from both, the behavioral difference is attributable to the $\beta_k \to \gamma_k$ pathway specifically. If tracked-only instead matches full precision modulation, the effect is likely mediated through belief quality.

3 Results

We report five sets of analyses. The first asks what the precision signal tracks. The second asks when that signal actually changes behavior, and whether the change comes from action sharpening rather than better beliefs. The third asks how the mechanism affects partner selection. The fourth and fifth use abrupt and gradual betrayal to identify the temporal boundary of the mechanism. We close with a compact parameter check on the precision pathway.

3.1 The precision signal tracks model fitness, not reward

The first question is whether β_k is mainly a reward-like estimate of partner value or a model-fitness estimate of predictive reliability. The trust game separates these quantities. A well-modeled cooperator and a well-modeled exploiter can be equally predictable, even though only one is consistently rewarding. This makes the reliability-versus-reward comparison a direct test of what the precision signal is measuring.

Across 30 seeds and 200 rounds per seed in the binary game, precision modulation yields no payoff advantage over the no-affect baseline: 534.6 vs. 542.1, mean difference -7.5, bootstrap CI $[-27.9,\ 13.8]$. The absence of a payoff benefit matters because it keeps the interpretation of β_k from collapsing into "the condition that earned more reward." Across partner-seed units, $|\mathrm{corr}(\beta_k, \mathrm{surprise})| = 0.701$ versus $|\mathrm{corr}(\beta_k, \mathrm{payoff})| = 0.419$. The seed-level reliability-over-reward margin is $+0.096$, CI $[0.027,\ 0.164]$. The precision signal is therefore more closely tied to whether the partner model predicts behavior well than to whether the partner is profitable.

The binary regime also explains why a reliable signal need not produce an immediate behavioral gain. The expected free energy difference between cooperating and defecting is large enough that the softmax policy posterior is already near-deterministic: β_k can move substantially without changing the selected action. The next section therefore asks whether the same signal becomes behaviorally consequential when the policy posterior is less saturated.

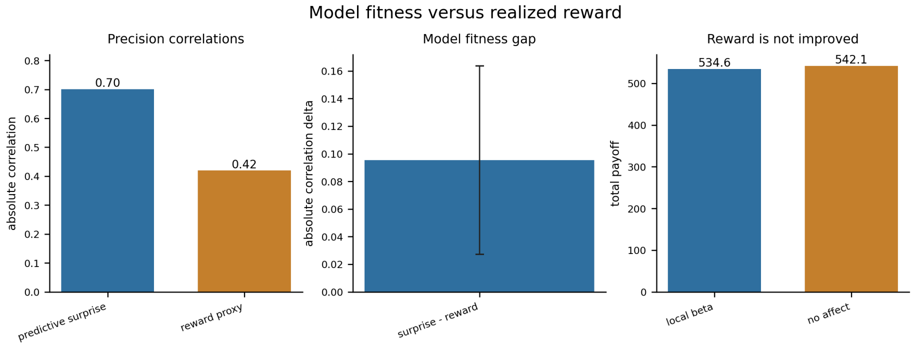

Fig. 2: Model-fitness analysis. Partner-local precision correlates more strongly with predictive surprise than with payoff across partner-seed units. Precision modulation yields no payoff advantage over the no-affect baseline in the binary regime, where policy entropy is already low.

3.2 Behavioral effect depends on policy entropy; lesion isolates the pathway

The second question is whether partner-local precision changes behavior only when there is room for precision to matter. In the binary game, sharpening or softening an already-peaked posterior changes little. In the graded investment game, the agent distributes probability mass across several investment levels; changing γ_k can move the selected investment level even when the underlying partner belief is unchanged.

We compare five seeds per variant, 200 rounds, across both regimes. In the binary game (Section 3.1) there is no payoff advantage and no meaningful entropy separation. In the graded game, the precision-modulated agent reaches total payoff 1884.6 versus 1859.4 for the no-affect baseline (+25.2), with mean policy entropy 7.94 versus 8.80 (-0.86). The same precision pathway that was behaviorally silent in the saturated binary game becomes visible once the task offers several plausible actions.

The tracked-only lesion addresses the pathway. In the graded regime the lesion—which tracks β_k fully but fixes $\gamma_k = \gamma_{\text{base}}$—has joint type-disposition belief accuracy 0.339 and stance accuracy 0.855, slightly better than the precision-modulated agent's 0.336 and 0.816. If the benefit came from better partner inference, the lesion should improve with the full model. It does not: its payoff and entropy match the no-affect baseline rather than the precision-modulated agent. The behavioral effect is therefore located at the action-selection stage, where β_k changes how sharply existing beliefs are expressed.

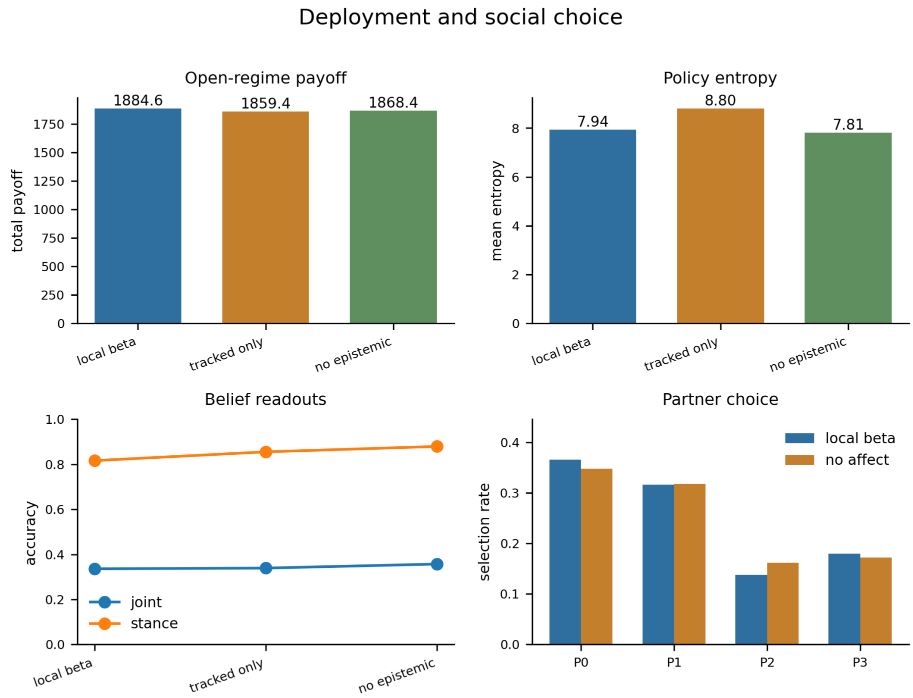

Fig. 3: Regime dependence and lesion comparison. Precision modulation improves payoff and reduces entropy in the decision-open graded regime but not in the near-saturated binary regime. The tracked-only lesion, which preserves belief accuracy while removing action-precision modulation, matches the no-affect baseline rather than the full-precision agent.

3.3 Partner selection reorganizes around model fitness

The third question is whether partner-local precision changes social allocation. In partner-choice regimes the agent expresses precision partly through whom it chooses to engage, not only how it acts once a partner is chosen. Aggregate payoff is a blunt measure here because selection shifts can occur before cumulative return separates.

Across five seeds and 200 rounds, aggregate payoff is nearly flat: 393.6 versus 393.2. The behavioral signature appears elsewhere. Policy entropy under precision modulation is 3.99 versus 4.83 without it, and selection rates shift directionally: the most predictable partner draws more engagement ($0.348 \rightarrow 0.366$) while a less predictable partner draws less ($0.162 \rightarrow 0.138$). The mechanism is not simply increasing reward; it is reorganizing interaction toward partners whose behavior the model can anticipate.

3.4 Abrupt betrayal reveals a boundary condition

The fourth question is what happens when a validated partner model suddenly becomes stale. Betrayal regimes introduce a stance switch mid-episode: one partner's behavior changes abruptly, creating a mismatch between the agent's accumulated model and the partner's new behavior. The question is not whether the agent can eventually adapt—it can—but what partner-local precision does during the transition.

After a stable pre-switch period, β_k for the switched partner reflects a model that has been repeatedly validated. When the partner changes, the agent enters the post-switch period with confidence inherited from the old regime. The possible failure mode is therefore temporal: precision can sharpen action before the partner-state posterior has reorganized.

Across 30 seeds and 120 rounds this is what we observe. Precision modulation yields lower whole-run payoff than the no-affect or tracked-only baselines: 1136.1 versus 1172.1, mean difference -36.1, bootstrap CI $[-63.7,\ -10.9]$, $p = 0.017$. Policy entropy is lower under precision modulation (8.38 vs. 8.74), so the loss is not caused by an inactive precision pathway. Post-switch, the agent returns to the switched partner less often (4.4 vs. 6.1 reencounters; selection rate 0.049 vs. 0.067) but is not better calibrated when it does: payoff per reencounter is 8.76 vs. 8.91 and wrong-type rate is 0.237 vs. 0.165. The mechanism made action more decisive without making the stale model correct.

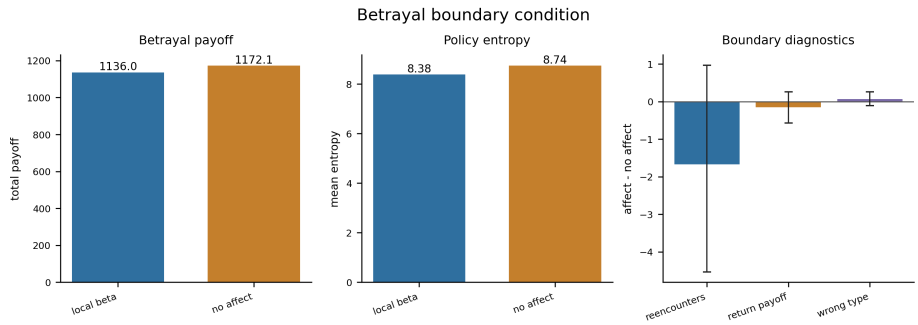

Fig. 4: Abrupt-betrayal boundary condition. Precision modulation reduces whole-run payoff, lowers policy entropy, and changes reallocation patterns without improving post-switch calibration. The agent acts more decisively on a model that is no longer valid.

3.5 Shock shape is the key moderator

The fifth question is whether the betrayal cost is caused by too much precision gain or by when that gain is deployed. A combined-caution variant raises β globally to reduce precision gain. If the problem were simply excessive confidence, this should help. It does not: combined caution lowers abrupt-betrayal payoff further (1115.0, $p = 0.003$, entropy 9.27). Blanket caution removes useful decisiveness in the non-betrayal portions of the episode without fixing the post-switch timing mismatch.

Gradual betrayal isolates the timing claim directly. When the partner's stance shifts slowly, the β_k posterior has time to track the changing prediction record before a large confidence surplus is applied to a mismatched model. Default precision modulation is neutral under gradual betrayal: payoff 1147.6 versus 1148.9 for no-affect, $p = 0.906$. Combined caution again reduces payoff (1118.3, $p = 0.005$). The cost of partner-local precision is therefore not an all-purpose penalty for being confident; it appears when behavioral change is faster than belief recalibration.

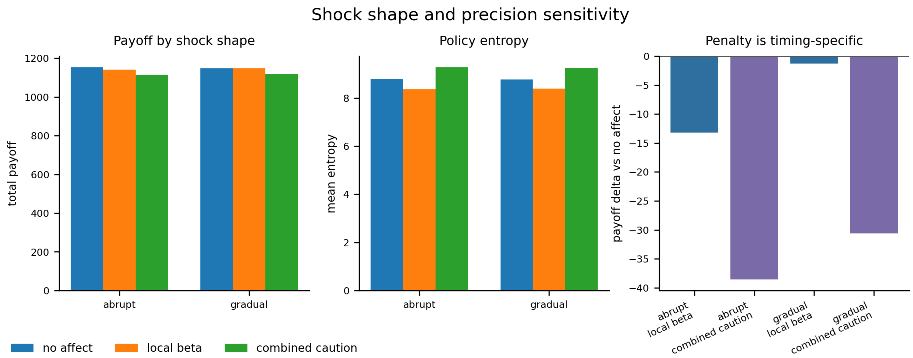

Fig. 5: Shock-shape sensitivity. Default local precision is neutral under gradual betrayal but harmful under abrupt betrayal. A generic caution variant reduces payoff in both regimes. The relevant boundary is the speed of behavioral change, not the magnitude of the precision response.

3.6 Precision-dynamics phenotypes

As a compact parameter check we verify that the β_k pathway responds to parameterization in an interpretable way before assigning behavioral meaning. In the open graded regime a low-gain variant produces narrow β_k movement (mean range 0.149 vs. 0.950 for default); high-gain and cautious-prior variants produce wider movement (1.318 and 1.435). The same ordering holds in betrayal. Payoff and action churn do not sort monotonically by beta-movement range, reinforcing the main result: beta dynamics are only meaningful in relation to the task regime in which they are deployed. These perturbations define computational phenotypes of the affective channel, not clinical models.

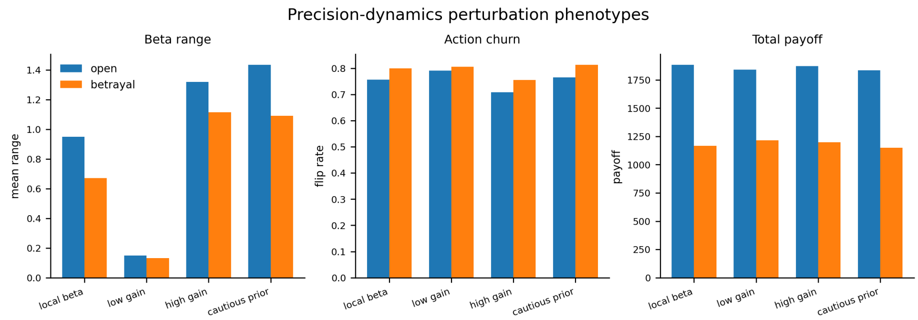

Fig. 6: Precision-dynamics phenotypes. Low-gain, high-gain, and cautious-prior variants separate clearly in β_k movement range. Behavioral consequences depend on task regime rather than beta magnitude alone.

4 Discussion

What the signal is. The main result is a dissociation between two components of social action: what the agent infers about a partner, and how confidently those inferences are expressed in behavior. The precision signal correlates more strongly with predictive reliability than with reward, arguing against a partner-value interpretation. The tracked-only lesion shows that the behavioral effect is not explained by better belief quality: the lesion maintains at-least-comparable partner beliefs while producing no improvement in payoff or entropy. Taken together, these results identify β_k as an estimate of how reliably the agent's partner model is supporting its choices—not a reflection of how valuable the partner is. A reliably defecting partner can carry high β_k (the model is working) alongside a low-value interaction; a high-value but unpredictable partner can carry low β_k (the model is not working) alongside a rewarding one. This distinction does not arise in models that treat precision as synonymous with experienced reward.

The timing problem is theoretically central. The abrupt-betrayal result is not merely a limitation—it is the clearest evidence that the mechanism is doing something structurally interesting. If β_k were behaviorally inert, the betrayal conditions would look identical across all variants: entropy flat, reallocation flat, calibration flat. Instead, precision modulation lowers entropy, changes reallocation, and makes action more decisive—while making it less accurate. The channel is active and consequential.

What this reveals about the architecture is the following. The Hesp et al. framework positions affective charge ϕ as a fast-moving signal (updated from each interaction's prediction error) that drives a slower-updating higher-order state β. In stable settings this is well-calibrated: many consistent outcomes drive β_k toward a reliable estimate. But when a partner changes abruptly, ϕ_k responds quickly to the new, surprising behavior while the partner-state posterior—which must integrate repeated evidence to revise its type-by-disposition beliefs—moves slowly. There is a window after the switch in which β_k still reflects pre-switch confidence while the lower-level model has not yet caught up. During this window, action sharpens around an inference that is becoming stale. Gradual change eliminates this mismatch: the posterior has time to track the behavioral shift before large confidence surplus accumulates. The boundary condition is therefore not about how much precision is gained but about how fast the social world changes relative to how fast beliefs reorganize.

Implications for social affect under volatility. This points toward a general design constraint for precision-based social affect. A fixed β_k update rate will be too fast under abrupt change (sharpen before beliefs catch up) and potentially too slow under gradual change (miss useful reliability signals). The natural extension is a volatility-sensitive update: when evidence suggests a partner's behavior is changing, weight recent ϕ_k signals less heavily, keeping β_k more diffuse until beliefs have had time to settle. This is structurally parallel to the volatility tracking in hierarchical Bayesian learning models [12, 1], where learning rates adapt to inferred environmental volatility. Applying this at the level of action precision—rather than belief updating—would allow the agent to respond to social change by temporarily widening its behavioral commitments rather than by becoming more decisive on unreliable grounds. Connecting these two levels of volatility tracking is an open problem.

The broader connection to social cognition research is worth noting. People adapt their social learning rates to partner volatility [1] and use distinct computational machinery for mentalizing that goes beyond standard reinforcement learning [8]. What we model here is a lower-level component: not reasoning about partner beliefs, but tracking one's own confidence in the model one uses to predict them [7]. These can dissociate. An agent that mentalizes accurately could still be miscalibrated about how much its own partner models should drive decisive action; partner-local β_k addresses the second question, not the first. A natural future direction is to combine the two: an agent that both models partner beliefs (as in theory-of-mind active inference [14, 15]) and maintains per-partner precision estimates over those models. Under social volatility, the precision layer would regulate how confidently the theory-of-mind layer's outputs are acted upon.

Limitations. Three features of the current model limit what can be claimed. First, the β_k tracker is an external auxiliary filter; in the Hesp et al. hierarchy, β is a proper hidden state over which variational inference is performed, allowing the agent to form prior expectations about its own future affective states and to act to preserve them. Giving β_k proper variational status would allow the agent to anticipate changes in partner reliability and plan around them—a richer form of proactive social regulation. Second, the partners are scripted: they follow predefined behavioral rules rather than being active-inference agents with their own social models. This isolates the focal agent's mechanism in a controlled way, but it cannot address whether partner-local precision would remain useful or informative when both parties are updating simultaneously. Third, the global-versus-local question remains open. Small follow-up runs with a single shared β show that global precision does not simply reproduce the local signal: local β_k retains a stronger association with predictive surprise, whereas global β can become weakly or even reward-associated. However, the first focused locality probe was mixed. Local β_k preserved the cleaner model-fitness signal, but global β achieved higher aggregate payoff in a five-seed mixed-partner task, and local β_k produced the strongest post-switch selection concentration. This means the present evidence supports partner-local precision as an interpretable mechanism, but not yet as a necessary one. The decisive test is whether a change in one partner contaminates policy entropy, selection, and payoff for untouched partners under global β more than under local β_k, in a design where the post-switch comparison is tightly controlled.

Future directions. The most important next experiment is a revised locality and interference diagnostic: hold several partners stable, switch one partner mid-run, and test separately whether local β_k preserves a cleaner reliability signal and whether that signal improves allocation after the shock. A second priority is a controlled shock-gradient study that varies how abruptly a partner's behavior changes while holding other parameters constant, allowing direct estimation of the timescale at which the betrayal cost first appears. Beyond these, the framework should be tested in environments with richer social structure—sequential exchange, reputational signaling, third-party information—where miscalibrated action confidence has more varied consequences. Eventually, testing against human behavioral data would allow partner-local β to function as a model of individual differences: people with higher β update rates should show more decisive post-cue reallocation, but also greater cost under abrupt social change.

5 Conclusion

Tracking predictive reliability is not the same as acting on it confidently. These are separable computational stages, and social affect in the active-inference framework can be understood as mediating the second one: a higher-order estimate of how well the agent's current model of a partner is earning the right to sharpen decisions. In repeated trust-game simulations, partner-local precision tracks reliability rather than reward, improves performance in high-entropy regimes through action sharpening rather than belief improvement, and reorganizes partner engagement before aggregate payoff separates. Under abrupt betrayal, the same mechanism causes harm—action becomes more decisive before beliefs have caught up with a changed partner. The value of calibrating action confidence to social model fitness is real and measurable, and its limits are informative about the temporal structure of adaptive social behavior.

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